[FAILED] Spec 29: Unconstrained Retention k=4

Zero Alpha Floor Ablation — k=4 from Scratch
Run: `k4_zero_retention_d512`

NL_Hecate Experiment Report

2026-03-17

Contents

1 Executive Summary **2**

2 Loss Trajectory **2**

3 Gradient Explosion **3**

4 Key Finding: Level Self-Organization (Steps 1000–3000) **3**
4.1 Alpha (Retention Gate) . . . 3
4.2 Theta (Write Strength) . . . 4

5 Divergence Analysis **4**
5.1 Root Cause: Two Fast-Forgetting Levels . . . 4
5.2 Why the Floor Prevented This . . . 5
5.3 Post-Divergence Gate Scrambling . . . 5

6 Eta (Momentum Gate) **5**

7 Throughput **6**

8 Implications **6**

9 Comparison with Other Runs **7**

1 Executive Summary

This run diverged catastrophically at step ∼3120. Loss exploded from ∼4.3 to >7.7 (worse than random init 11.0 would suggest recovery is impossible), with gradient norms reaching 10^{17}. The run was killed at step ∼4300.

However, the first 3000 steps produced the key finding of this experiment: CMS levels self-organized their retention rates by frequency without any floor constraint. This validated the hypothesis that `alpha_floor=0.8` was preventing slow levels from finding their natural equilibrium. The instability at k=4 appears to be caused by two fast-forgetting levels (L0 + L1) interfering with each other, not by the zero floor per se — the parallel k=2 zero-floor run sailed through the same step range without issues.

Parameter	**Value**
Model	d=512, 8 heads, 4 blocks, k=4
CMS chunk sizes	[1, 8, 64, 512]
Tape strategies	[exact, proxy, proxy, proxy]
Alpha floor	[0.0, 0.0, 0.0, 0.0]
Theta ceil	[1.0, 1.0, 1.0, 1.0]
LR	3e-4, warmup 500 steps
Data	Dolmino-100B
Stacked params	76,687,440 (57.8M base)
Throughput	∼520 tok/s
GPU	A6000 (GPU 0)
Steps completed	∼4300 / 100,000 planned
Outcome	**DIVERGED at step ∼3120**

Table 1: Run configuration summary.

2 Loss Trajectory

The model trained normally for the first ∼3000 steps, reaching loss ∼4.3 (comparable to the k=2 zero-floor control at the same step count). At step ∼3120, gradient norms began escalating exponentially, and by step ∼3200 the loss had jumped to >7.5.

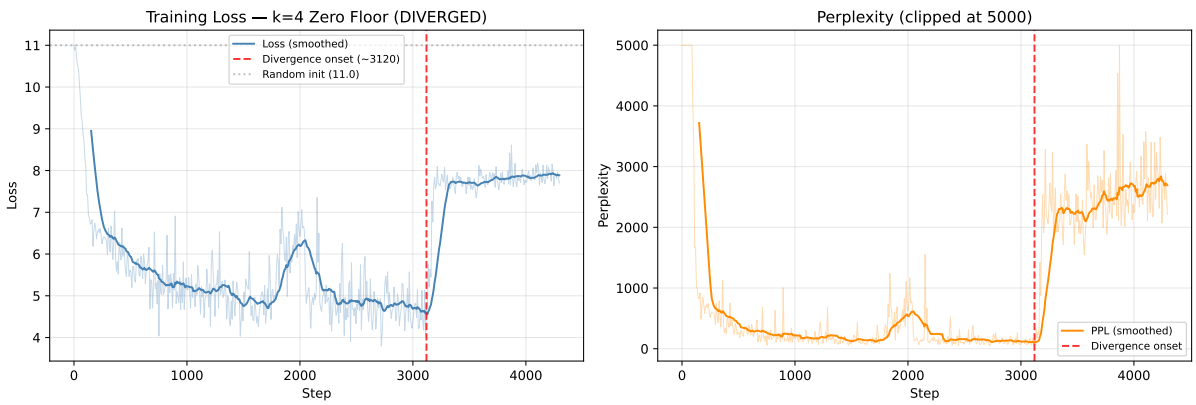

Figure 1: Loss and perplexity. Red dashed line marks divergence onset at step ∼3120.

3 Gradient Explosion

The divergence followed a clear exponential pattern:

Step	Grad Norm	Status
3088	2.1	Healthy
3096	510	First spike
3120	1,500	Escalating
3128	360,000	Point of no return
3136	11,000,000	Catastrophic
3200	2.7×10^{13}	Fully diverged
4000	$\sim 10^{17}$	Saturated

Table 2: Gradient norm escalation through the divergence event.

Note: `max_grad_norm=1.0` clipping was active but operates on the *outer* gradient norm. The explosion originates inside the memory update dynamics before outer gradient clipping can intervene.

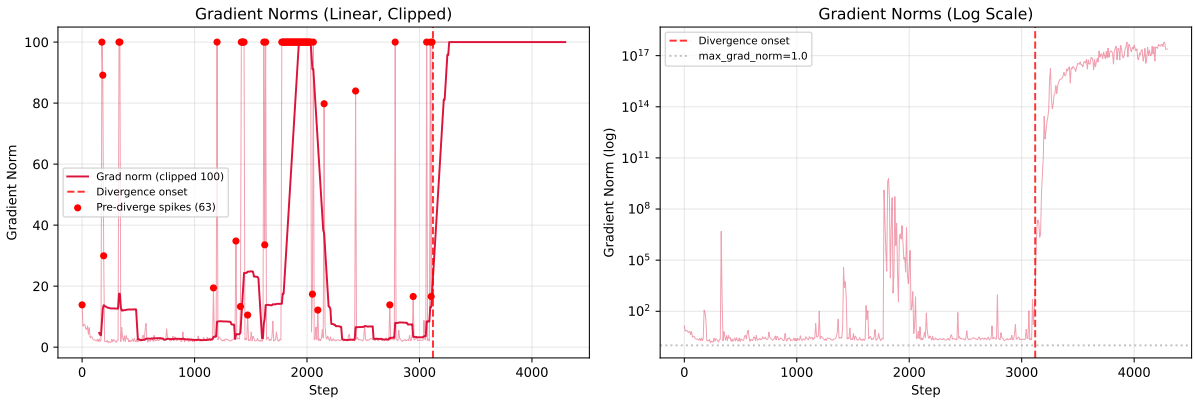

Figure 2: Gradient norms on linear (clipped at 100) and log scales.

4 Key Finding: Level Self-Organization (Steps 1000–3000)

Before divergence, the tape summaries at steps 1000, 2000, and 3000 showed exactly the level differentiation predicted by the spec 29 hypothesis.

4.1 Alpha (Retention Gate)

Level	Step 1000	Step 2000	Step 3000	Step 4000 (diverged)
L0 (every token)	0.749	0.691	0.737	0.921
L1 (every 8)	0.919	0.752	0.673	0.758
L2 (every 64)	0.988	0.988	0.988	0.988
L3 (every 512)	0.993	0.993	0.993	0.991

Table 3: Alpha (retention) per level across tape summaries. L1 dropped to 0.673 by step 3000 — well below the old 0.8 floor. Post-divergence (step 4000), L0 alpha *increased* to 0.921, indicating gate dynamics were scrambled.

Key observations (pre-divergence):

- **Monotonic alpha gradient**: L0 (0.74) < L1 (0.67) < L2 (0.99) < L3 (0.99). Fast levels forget more, slow levels retain.
- **L0 below old floor**: At 0.74, L0 chose faster forgetting than the 0.8 floor would have allowed.
- **L1 dropped dramatically**: From 0.92 → 0.67 in 2000 steps. This level was actively adapting its retention.
- **L2/L3 stable**: Slow levels maintained near-1.0 retention — frequency-appropriate.

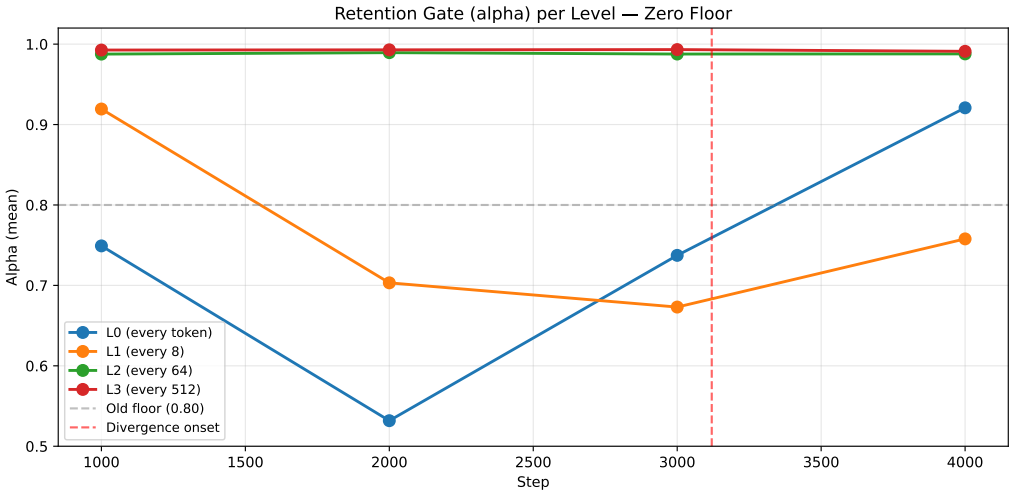

Figure 3: Alpha evolution per level. The old floor (0.8) is shown for reference. L0 and L1 both dropped below it.

4.2 Theta (Write Strength)

Level	Step 1000	Step 2000	Step 3000	Step 4000
L0	0.193	0.333	0.388	0.642
L1	0.013	0.137	0.212	0.038
L2	0.002	0.002	0.002	0.002
L3	0.0005	0.0005	0.0005	0.0007

Table 4: Theta (write strength) per level. L0 and L1 were both actively writing pre-divergence. Post-divergence, L1 theta collapsed back to 0.038.

5 Divergence Analysis

5.1 Root Cause: Two Fast-Forgetting Levels

The k=2 zero-floor run (identical config except k=2) passed through the same step range (3000–3500) with completely stable gradient norms (∼2.0). This rules out the dataset as a cause. The instability is k=4-specific.

At step 3000, both L0 and L1 were in aggressive forgetting + writing regimes:

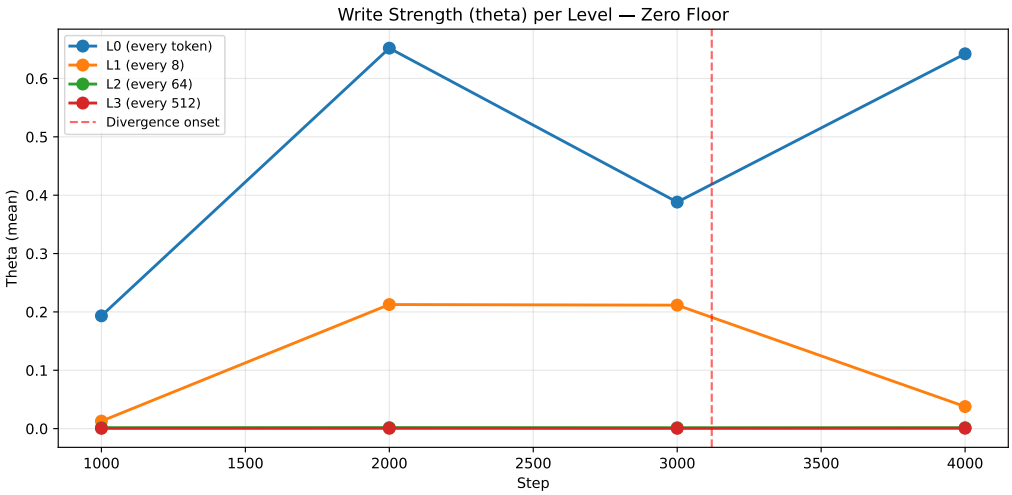

Figure 4: Theta evolution per level.

- **L0**: $\alpha = 0.74$, $\theta = 0.39$ — forgetting 26% per step, writing moderately
- **L1**: $\alpha = 0.67$, $\theta = 0.21$ — forgetting 33% per fire, writing actively

Both levels feed into the same MAG composition. When L1's aggressive writes interfere with L0's attention pathway, a feedback loop can form:

1. L1 writes stale or noisy content into memory
2. MAG composition mixes L1's output with attention, corrupting the residual stream
3. Attention receives corrupted input, producing larger gradients
4. Larger gradients cause L0 to write even more aggressively
5. Repeat until explosion

5.2 Why the Floor Prevented This

With `alpha_floor=0.8`, L1 could only forget 20% per fire and was effectively dormant ($\theta < 0.05$). This prevented the dual-fast-forgetting instability because L1 couldn't write aggressively enough to interfere with L0.

5.3 Post-Divergence Gate Scrambling

After divergence (step 4000), L0 alpha jumped from 0.74 to 0.92 (high retention) while L1 theta collapsed from 0.21 to 0.04 (dormant). The gradient explosion reset the gates to a conservative, low-activity state — but the damage was already done.

6 Eta (Momentum Gate)

All levels maintained healthy eta values throughout, including post-divergence. The momentum gate was not a factor in the instability.

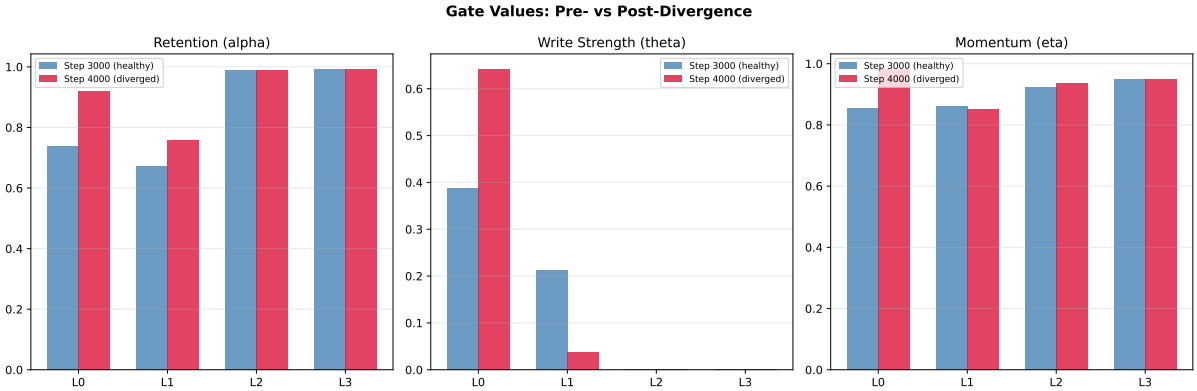

Figure 5: Gate values pre-divergence (step 3000) vs post-divergence (step 4000).

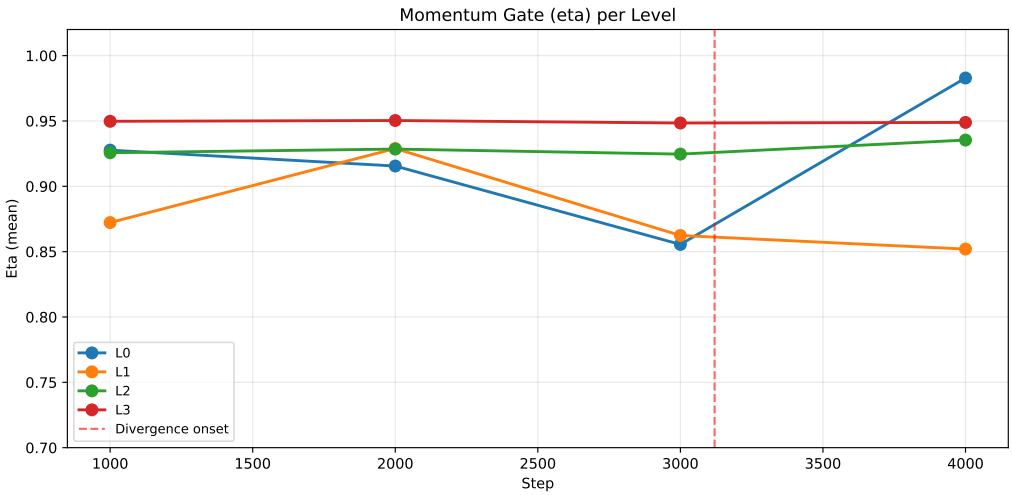

Figure 6: Eta remained stable across all levels.

7 Throughput

8 Implications

1. **Zero floor IS viable for level differentiation.** Steps 1000–3000 showed exactly the frequency-appropriate retention self-organization predicted by spec 29. The hypothesis is validated even though the run failed.

2. **k=4 from scratch with zero floor is unstable.** Two fast levels (L0 + L1) with unconstrained forgetting can enter a destructive feedback loop. This does NOT mean the floor is necessary for all levels — it means k=4 requires either:

   - A per-level floor: e.g., L0=0.0, L1=0.5, L2=0.0, L3=0.0
   - Better gradient clipping inside the memory update
   - Graduated introduction of levels (push-up with zero floor)

3. **k=2 zero floor is the right next step.** The parallel k=2 run is stable. Master k=2 before attempting k=3 or k=4.

4. **Future k=3/k=4 attempts should consider per-level floors** scaled to prevent the dual-fast-forgetting instability while allowing slow levels full freedom.

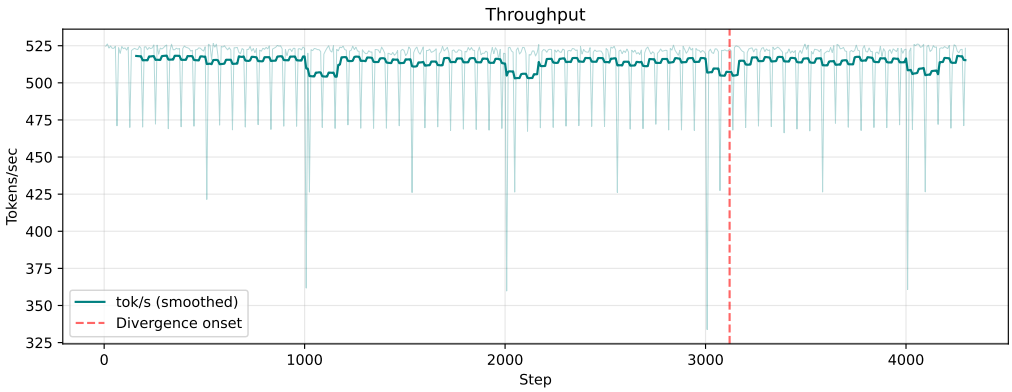

Figure 7: Throughput held steady at ∼520 tok/s throughout the run.

9 Comparison with Other Runs

Run	k	Floor	Steps	Loss	Status
spec25_30k	2	0.8	30K	2.75	Complete
spec27_130k	2	0.8	130K	2.60	Complete
pushup_k3	3	0.8	95K	2.53 (best)	Complete (L2 dormant)
This run	**4**	**0.0**	**4.3K**	**4.3 → 7.7**	**DIVERGED**
k2_zero_floor	2	0.0	running	∼4.2 @ 3K	In progress

Table 5: Comparison across experiment lineage.